

R Code

```
#install.packages("brms", dependencies = TRUE)
#install.packages("truncnorm", dependencies = TRUE)
#library(brms)
#library(dplyr)
#library(ggplot2)
#library(truncnorm)

#Part 1

df_powerpose <- read.table("C:/Users/Aaditya/Downloads/df_powerpose.csv", header=TRUE, sep=",")
head(df_powerpose)

#Modify the data
df_powerpose$y <- rowMeans(df_powerpose[, c("testm1", "testm2")])
df_powerpose$hptreat <- ifelse(df_powerpose$hptreat == "High", 1, 0)
df_powerpose$female <- ifelse(df_powerpose$female == "Male", 1, 0)
head(df_powerpose)
print(df_powerpose$X)

#Fit the data

fit <- brm(
  y ~ hptreat + female + age + X,
  data = df_powerpose,
  sample_prior = "only",
  chains = 4,
  iter = 2000,
  seed = 42
)

#I will have to come up with weak priors for b.

priors <- c(
  prior(normal(0, 5), class = "b"),
  prior(student_t(3, 0, 10), class = "Intercept"),
  prior(student_t(3, 0, 10), class = "sigma")
)

fit <- brm(
  y ~ hptreat + female + age + X,
  data = df_powerpose,
  prior = priors,
  sample_prior = "only",
  chains = 4,
  iter = 2000,
  seed = 42
)
pp_check(fit)

#Better priors can be set.

priors <- c(
  prior(normal(0, 5), class = "b"),
  prior(normal(50, 10), class = "Intercept"),
  prior(student_t(3, 0, 10), class = "sigma")
)

fit <- brm(
```

```

y ~ hptreat + female + age + X,
data = df_powerpose,
prior = priors,
sample_prior = "only",
chains = 4,
iter = 2000,
seed = 42
)

pp_check(fit)

#Attempt at fixing priors

priors <- c(
  prior(normal(0, 2), class = "b"),
  prior(normal(50, 10), class = "Intercept"),
  prior(exponential(1.5), class = "sigma")
)

fit <- brm(
  y ~ hptreat + female + age + X,
  data = df_powerpose,
  prior = priors,
  sample_prior = "only",
  chains = 4,
  iter = 2000,
  seed = 42
)

#Extract coefficient of hptreat
#This would give us information regarding our claim
summary(fit)

#coefficient of hptreat = -0.02 with confidence intervals = [-3.87,3.84]
#This does not give us a strong evidence supporting our hypothesis.
#If the value of hptreat was negative with narrower confidence intervals
#surrounding our value, we could safely accept the hypothesis. Lack of strong
#evidence suggests that the hypothesis cannot be accepted.

#Problem-2
#Exercise-2.1
number_of_crossings <- function(alpha, beta, length){
  lambda <- exp(alpha + beta * length)
  N <- rpois(1, lambda = lambda)
  return(N)
}

#Exercise-2.2
#alpha ~ truncated Normal(0.15,0.1)
#beta ~ truncated Normal(0.25,0.05)

length <- 4
n_samples <- 100
install.packages("truncnorm")
library(truncnorm)
alpha <- rtruncnorm(n_samples, a = 0, b = Inf, mean = 0.15, sd = 0.1)
beta <- rtruncnorm(n_samples, a = 0, b = Inf, mean = 0.25, sd = 0.05)
N <- numeric(length)
for (i in 1:n_samples){
  N[i] <- number_of_crossings(alpha[i], beta[i], length)
}

```

```
print(N)
```

```
#Exercise-2.3
```

```
crossings <- read.table("Ass-5/crossings.csv", header = TRUE, sep = ",")
```

```
head(crossings)
```

```
#Fitting model-M1
```

```
#Assumption- Alpha remains same in both German and English
```

```
library(brms)
```

```
#using priors from previous question.
```

```
priors <- c(  
  prior(normal(0.15, 0.1), class = "Intercept"),  
  prior(normal(0.25, 0.05), class = "b")  
)
```

```
fit_m1 <- brm(  
  formula = nCross ~ s.length,  
  family = poisson(link = "log"),  
  data = crossings,  
  prior = priors,  
  chains = 4,  
  iter = 2000,  
  seed = 42  
)
```

```
pp_check(fit_m1)
```

```
summary(fit_m1)
```

```
#coefficients: Intercept = -1.46 and cI = [-1.57,-1.34]
```

```
#s.length = 0.15 and cI = [0.14, 0.16]
```

```
#Fitting model-M1
```

```
#Assumption- Beta values differ for English and German with increase in length
```

```
#Need to change data
```

```
crossings$Language <- ifelse(crossings$Language == "German", 1, 0)
```

```
head(crossings)
```

```
priors <- c(  
  prior(normal(0.15, 0.1), class = "Intercept"),  
  prior(normal(0.25, 0.05), class = "b", coef = "s.length"),  
  prior(normal(0, 0.15), class = "b", coef = "Language"),  
  prior(normal(0, 0.15), class = "b", coef = "s.length:Language")  
)
```

```
fit_m2 <- brm(  
  formula = nCross ~ s.length * Language,  
  family = poisson(link = "log"),  
  data = crossings,  
  prior = priors,  
  chains = 4,  
  iter = 2000,  
  seed = 42  
)
```

```
pp_check(fit_m2)
```

```
summary(fit_m2)
```

```
#coefficients: Intercept = -1.12 and cI = [-1.27,-0.97]
```

```
#s.length = 0.15 and cI = [0.10, 0.12]
```

```
#Language = -0.6 and cI = [-0.79, -0.41]
```

```
#s.Length:Language = 0.07 and cI = [0.06, 0.08]
```

```
#Exercise -2.4
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
observed <- read.table("Ass-5/crossings.csv", sep = ",", header = TRUE)
```

```

# Visualize average rate of crossings
observed %>% group_by(Language, s.length) %>%
  summarise(mean.crossings=mean(nCross)) %>%
  ggplot(aes(x = s.length, y = mean.crossings,
             group = Language, color = Language)) +
  geom_point() + geom_line()
# Code/center the predictors

observed$s.length <- observed$s.length - mean(observed$s.length)
observed$lang <- ifelse(observed$Language == "German", 1, 0)
# These two vectors will store log predictive densities
# in each fold
lpds.m1 <- c()
lpds.m2 <- c()

untested <- observed
library(brms)

for (k in 1:5){

# Prepare test data and training data

ytest <- sample_n(untested, size = nrow(observed) / 5)
ytrain <- setdiff(observed, ytest)
untested <- setdiff(untested, ytest)
# Fit the models M1 and M2 on training data
fit.m1 <-
  brm(nCross ~ 1 + s.length, data = ytrain,
      family = poisson(link = "log"),
      prior = c(prior(normal(0.15, 0.1), class = Intercept),
                prior(normal(0, 0.15), class = b)),
      cores = 4)

fit.m2 <-
  brm(nCross ~ 1 + s.length + lang + s.length * lang,
      data = ytrain,
      family = poisson(link = "log"),
      prior = c(prior(normal(0.15, 0.1), class = Intercept),
                prior(normal(0, 0.15), class = b)),
      cores = 4)

post.m1 <- posterior_samples(fit.m1)
post.m2 <- posterior_samples(fit.m2)
# Calculate log point-wise predictive density using test data
lppd.m1 <- 0
lppd.m2 <- 0
for (i in 1:nrow(ytest)){
  lpd_im1 <- log(mean(dpois(ytest[i, ]$nCross,
                           lambda = exp(post.m1[, 1] +
                                         post.m1[, 2]*ytest[i, ]$s.length)))) #
  nolint: line_length_linter.
  lppd.m1 <- lppd.m1 + lpd_im1
  lpd_im2 <- log(mean(dpois(ytest[i, ]$nCross,
                           lambda = exp(post.m2[, 1] +
                                         post.m2[, 2] * ytest[i, ]$s.length +
                                         post.m2[, 3] * ytest[i, ]$lang +
                                         post.m2[, 4] * ytest[i, ]$s.length * ytest[i, ]$lang)
                           # nolint: line_length_linter.
                           )))
  lppd.m2 <- lppd.m2 + lpd_im2
}
lpds.m1 <- c(lpds.m1, lppd.m1)

```

```
}
```

```
# Predictive accuracy of model M1
elpd.m1 <- sum(lpds.m1)
# Predictive accuracy of model M2
elpd.m2 <- sum(lpds.m2)
# Evidence in favor of M2 over M1
difference_elpd <- elpd.m2 - elpd.m1
print(difference_elpd)

#With an expected log predictive density (ELPD) difference of 2239.29,
#Model M2 significantly outperforms Model M1. This substantial margin provides
#compelling evidence that the rate of crossing dependencies is driven by both
#sentence length and language, demonstrating distinct growth trajectories for
#English and German.
```











